

CSE 4203-4.2 Integer representations + Algorithms.

* Addition Algorithm:

$$-a = (1110)_2 \quad n=4$$

$$-b = (1011)_2 \quad i=0 \dots (n-1)$$

$$\begin{array}{r} 1110 \\ + 1011 \\ \hline (11001)_2 \end{array}$$

$$a_n + b_n + c_{n-1} = \frac{c_n \cdot 2}{1} + S_n$$

$$a_0 + b_0 + 0 = c_0 \cdot 2 + S_0$$

$$0 + 1 + 0 = 0 \cdot 2 + \underline{1}$$

$$a_1 + b_1 + c_0 = c_1 \cdot 2 + S_1$$

$$1 + 1 + 0 = 1 \cdot 2 + \underline{0}$$

$$a_2 + b_2 + c_1 = c_2 \cdot 2 + S_2$$

$$1 + 0 + 1 = 1 \cdot 2 + \underline{0}$$

$$a_3 + b_3 + c_2 = \frac{c_3}{S_4} \cdot 2 + S_3$$

$$1 + 1 + 1 = \underline{1} \cdot 2 + \underline{1}$$

Sum

$$\begin{cases} c = 0 \\ \text{for } i = 0 \dots (n-1) \{ \\ \quad d = \lfloor (a_i + b_i + c) / 2 \rfloor \\ \quad S_i = a_i + b_i + c - 2 \cdot d \\ \quad c = d \} \\ S_n = c \end{cases}$$

$$d = \lfloor (0 + 1 + 0) / 2 \rfloor = \lfloor 1/2 \rfloor = 0$$

$$S_0 = 0 + 1 + 0 - 2 \cdot 0 = 1$$

$$c = d = 0$$

$$d = \lfloor (1 + 1 + 0) / 2 \rfloor = \lfloor 1 \rfloor = 1$$

$$S_1 = (1 + 1 + 0) - 2 \cdot 1 = 0$$

$$c = d = 1$$

*) Multiplication Algorithm:

$$ab = a(b_0 \cdot 2^0 + b_1 \cdot 2^1 + b_2 \cdot 2^2 + \dots + b_{n-1} \cdot 2^{n-1})$$

$$= ab_0 \cdot 2^0 + ab_1 \cdot 2^1 + ab_2 \cdot 2^2 + \dots + ab_{n-1} \cdot 2^{n-1} //$$

multiplying each time by 2, shift the expansion 1-digit to the left and add a zero on the right.

$$\left\{ \begin{array}{l} a = (110)_2 \quad b = (101)_2 \\ ab_0 2^0 = (110)_2 \cdot 1 \cdot 2^0 = (110)_2 - \\ ab_1 2^1 = (110)_2 \cdot 0 \cdot 2^1 = (\underline{0000})_2 - \\ ab_2 2^2 = \overbrace{(110)_2}^{\text{shift left 2}} \cdot 1 \cdot 2^2 = (\underline{11000})_2 - \end{array} \right.$$

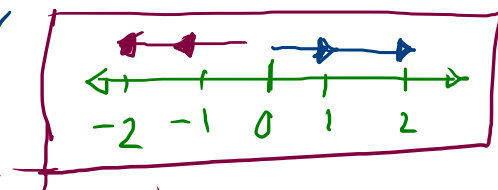
$$\begin{array}{r} 110 \\ 0000 \\ 11000 \\ \hline (11110)_2 \end{array}$$

$$\left. \begin{array}{r} 110 = \\ \times 101 \\ \hline 110 \\ \rightarrow 000 \times = \\ 110 \times \underline{\underline{x}} = \\ \hline (11110)_2 \end{array} \right\} \rightarrow$$

⑩ Division

$$a, d \in \mathbb{Z}, d > 0 \rightarrow d \nmid a$$

↓
divisor
↓
dividend



$$\begin{array}{r} 3 \overline{) \begin{array}{r} -11 \\ -12 \\ \hline 1 \end{array}} (-4 \end{array}$$

$q = 0$
 $r = |a|$
 while $r \geq d$
 $r = r - d$
 $q = q + 1$
 if $a < 0$ and $r > 0$ then
 $r = d - r$
 $q = -(q + 1)$

$$q = a \text{ div } d = d \mid a$$

$$r = a \bmod d = a \% d$$

$$a = -11 \quad d = 3$$

$$q = 0, r = |-11| = 11$$

while

$$r > d \rightarrow 11 > 3? \checkmark \rightarrow r = 11 - 3 = 8, q = 1$$

$$r > d \rightarrow 8 > 3? \checkmark \rightarrow r = 8 - 3 = 5, q = 2$$

$$r > d \rightarrow 5 > 3? \checkmark \rightarrow r = 5 - 3 = 2, q = 3$$

$$r > d \rightarrow 2 > 3? \text{X}$$

exit

$$\left. \begin{array}{l} a < 0 \rightarrow -11 < 0 \checkmark \\ r > 0 \rightarrow 2 > 0 \checkmark \end{array} \right\} \begin{array}{l} r = 3 - 2 = \underline{1} \\ q = -(3 + 1) = -4 \end{array}$$